Calxeda shuts down
Calxeda, the ARM server chip pioneer has shut down due to additional financial problems. Read the full story: **http://dgit.in/PoorCalxeda**

Sony Dualshock 4
Want to know what's new and what's old with Sony's DualShock 4 controllers? **http://dgit.in/18Tcko5**

AMD R9 290X v/s NVIDIA GTX 780 Ti

A multi-faceted look at the flagships from both NVIDIA and AMD

Mithun Mohandas
readersletters@thinkdigit.com

Now that we have the flagships from both AMD as well as NVIDIA, it is high time to get a comparison out before we get the dual-GPU flagships from both manufacturers if there ever will be one. While the GTX 780 was touted to be the flagship, AMD's introduction of the R9 290X tilted the scales in the favour of AMD which is why NVIDIA came out with the GTX 780 Ti. While the pricing schema has remained the same with the NVIDIA card being priced way above the AMD card, the performance trend has changed for the better. AMD is no longer lagging far behind and that coupled with the fact that the card is cheaper makes the AMD seem a much sweeter deal but that all comes down to where your priorities lie. Read on to figure out which card suits your definition of what a graphics card should do.

Features

Mantle and DirectX

The AMD card has support for the new Mantle API which is due to be released in an update this December. The new API allows direct access to the onboard GDDR5 memory so game developers may be able to make better use of that memory for not only storing frames but also use up memory for loading maps. While this hasn't been implemented and there is a lot to consider before going for such an approach (like ensuring that performance doesn't take a hit while using the onboard memory), it offers diverse possibilities and has a lot of potential. Battlefield 4 is going to be the first game where we'll be able to see the new API in action. NVIDIA hasn't mentioned anything about Mantle so far and it's not as if only the R9 290 and the R9 290X are capable of using Mantle. Previous generation AMD cards will be able to render the Mantle API as well.

AMD 1 - NVIDIA 0

Raptr GeForce Experience and Shadowplay

The most recent addition to the NVIDIA environment would be the GeForce Experience widget which came along with the recent driver package. And close on the heels of the same, AMD launched a similar feature by incorporating Raptr's client into their drivers. These two software collect information regarding your computer configuration and monitors your FPS. After a while this data is sent to their servers. By analysing this information they can come up with the optimal configuration where you can get the most FPS at the highest settings possible without compromising on frame rate. The implementation of the system by both manufacturers needs to be looked at. NVIDIA's tool is created solely by them and is less intrusive compared to Raptr's client which pops up after each gaming session. However, Raptr has a better API which recognises games and incorporates an UI overlay similar to that of steam. Also, the whole community aspect comes into the picture with Raptr while GeForce Experience doesn't have that.

Another new feature that has been included is Shadowplay which allows you to save gameplay videos without having to utilise an external capture device. The feature is limited to cards that came after the GTX 660.

AMD 1 - NVIDIA 2

Memory

The AMD card comes with 4GBs of GDDR5 memory and the NVIDIA card has 3GBs of GDDR5 memory but that's not where the story ends. AMD's R9 290X offers a memory bandwidth of 320GB/s with a bus width of 512bit while NVIDIA's GTX 780 Ti gives you 336GB/s with a bus width of 384bit. The overall performance comes from the overall bandwidth and this is where the NVIDIA card gets ahead by a narrow margin. This marginal increment comes into play when you have memory intensive tasks like Anti-Aliasing, HDR